
EGC 301P-Operating Systems Lab

Mini Project – Report

Problem Statement

The project aims to develop a Course Registration (Academia) Portal that is user-friendly and multifunctional.

- There are 3 main actors: Admin, Faculty and Student.
- Account holders have to pass through a login system to enter their accounts, and all these User accounts will be managed by the Administrator.
- The application should possess password-protected administrative access, thus preventing the whole management system from unauthorized access.

Implementation details

Follow these steps to implement and run the project:

- Extract the folder and use terminal in the main directory
- Compile the files in the main directory using these commands:
 - `gcc server.c admin.c faculty.c student.c utils.c -o server -pthread`
 - `gcc client.c -o client`

Now open one terminal for client and one for server window and run:

- `./server` (in server window)
- `./client` (in client window)

Now, the user can use these credentials testing the program:

Admin: user – admin1, password – pass1

user – admin2, password – pass2

Similarly, [stud1, pass1], [stud2, pass2] etc. for student and [prof1, pass1], [prof2, pass2] etc, for faculty. You can also refer to the text files in the [data] sub-folder in the main directory.

Brief Explanations of Source Codes

client.c

Implements a TCP client that connects to the server, handles user login, and sends commands for course-related operations. Displays server responses and guides users through their role-based menus and actions.

server.c

Runs a multithreaded TCP server that handles multiple client connections and authenticates users based on role. Routes each client to appropriate role-specific functions (admin, faculty, student) with secure file access.

student.c

Handles student functionalities like viewing, enrolling in, and dropping courses, as well as changing passwords. Ensures safe file access with locking for enrollment updates.

utils.c

Offers utility functions for prompting users, validating IDs, checking enrollments, and formatting output. Used across all modules for consistent input validation and display formatting.

admin.c

Enables admins to manage student and faculty records—add, modify, view, activate, or block students. Uses file locking to maintain data consistency during admin operations.

faculty.c

Provides faculty options to manage courses (add/update/delete/view) and change account passwords. Applies file locking during course updates to avoid data conflicts.

Output Screenshots

1) Admin Interface: Viewing Student and Faculty Info

```
***** Welcome Back to Academia :: Course Registration *****
Login Type
Enter Your Choice { 1. Admin, 2. Professor, 3. Student } : 1
Enter Your ID: admin1
Enter Your Password: pass1
***** Welcome to ADMIN MENU *****
1. ADD Student
2. VIEW Student Details
3. ADD Faculty
4. VIEW Faculty Details
5. ACTIVATE Student
6. BLOCK Student
7. MODIFY Student Details
8. MODIFY Faculty Details
9. LOGOUT and EXIT
ENTER YOUR CHOICE: 2
+-----+-----+-----+-----+-----+
| ID      | Name      | Email      | Phone      | Status    |
+-----+-----+-----+-----+-----+
| stud1   | John      | john@example.com | 1234567890 | Active    |
| stud2   | Alice     | alice@example.com | 2345678901 | Active    |
| stud3   | Bob       | bob@example.com   | 3456789012 | Blocked   |
+-----+-----+-----+-----+-----+

***** Welcome to ADMIN MENU *****
1. ADD Student
2. VIEW Student Details
3. ADD Faculty
4. VIEW Faculty Details
5. ACTIVATE Student
6. BLOCK Student
7. MODIFY Student Details
8. MODIFY Faculty Details
9. LOGOUT and EXIT
ENTER YOUR CHOICE: 4
+-----+-----+-----+-----+-----+
| ID      | Name      | Email      | Phone      | Password   |
+-----+-----+-----+-----+-----+
| prof1   | Jane      | jane@example.com | 0987654321 | pass1     |
| prof2   | Mike      | mike@example.com | 9876543210 | pass2     |
+-----+-----+-----+-----+-----+
```

Server-Side Response:

```
Server listening on port 8080...
New client connected
Client authenticated: admin1 (Role: 1)
```

2) Student Interface: Viewing Offered and Enrolled Courses

```
Enter Your ID: stud2
Enter Your Password: pass2
***** Welcome to STUDENT MENU *****
1. VIEW ALL Courses
2. ENROLL (pick) NEW Course
3. DROP Course
4. VIEW ENROLLED Course Details
5. CHANGE PASSWORD
6. LOGOUT and EXIT
ENTER YOUR CHOICE: 1
+-----+-----+-----+-----+
| Course ID | Name           | Seats | Faculty ID |
+-----+-----+-----+-----+
| CS101     | IntroToCs      | 30    | prof1      |
| MATH201   | Calculus       | 35    | prof1      |
| ENG301    | Literature     | 19    | prof2      |
+-----+-----+-----+-----+

***** Welcome to STUDENT MENU *****
1. VIEW ALL Courses
2. ENROLL (pick) NEW Course
3. DROP Course
4. VIEW ENROLLED Course Details
5. CHANGE PASSWORD
6. LOGOUT and EXIT
ENTER YOUR CHOICE: 4
+-----+-----+-----+-----+
| Course ID | Name           | Seats | Faculty ID |
+-----+-----+-----+-----+
| CS101     | IntroToCs      | 30    | prof1      |
| ENG301    | Literature     | 19    | prof2      |
+-----+-----+-----+-----+

***** Welcome to STUDENT MENU *****
1. VIEW ALL Courses
2. ENROLL (pick) NEW Course
3. DROP Course
4. VIEW ENROLLED Course Details
5. CHANGE PASSWORD
6. LOGOUT and EXIT
ENTER YOUR CHOICE: 6
Logging out...
Disconnected from server
```

Server-Side Response

```
root@DESKTOP-R12LH4E:~/miniProj/OS-MiniProject# ./server
Server listening on port 8080...
New client connected
Client authenticated: stud2 (Role: 3)
Client logged out: stud2
|
```

3) Faculty Interface: Adding New Course to The Catalog

```
Enter Your ID: prof1
Enter Your Password: pass1
***** Welcome to FACULTY MENU *****
1. VIEW OFFERING Courses
2. ADD NEW Course
3. REMOVE Course from Catalog
4. UPDATE Course Details
5. CHANGE PASSWORD
6. LOGOUT and EXIT
ENTER YOUR CHOICE: 1
+-----+-----+-----+-----+
| Course ID | Name           | Seats | Faculty ID |
+-----+-----+-----+-----+
| CS101     | IntroToCs      | 30    | prof1      |
| MATH201   | Calculus       | 35    | prof1      |
+-----+-----+-----+-----+

***** Welcome to FACULTY MENU *****
1. VIEW OFFERING Courses
2. ADD NEW Course
3. REMOVE Course from Catalog
4. UPDATE Course Details
5. CHANGE PASSWORD
6. LOGOUT and EXIT
ENTER YOUR CHOICE: 2
Enter Course ID: EGC301P
Enter Course Name: OS_Lab
Enter Number of Seats: 192
Course added successfully.
***** Welcome to FACULTY MENU *****
1. VIEW OFFERING Courses
2. ADD NEW Course
3. REMOVE Course from Catalog
4. UPDATE Course Details
5. CHANGE PASSWORD
6. LOGOUT and EXIT
ENTER YOUR CHOICE: 1
+-----+-----+-----+-----+
| Course ID | Name           | Seats | Faculty ID |
+-----+-----+-----+-----+
| CS101     | IntroToCs      | 30    | prof1      |
| MATH201   | Calculus       | 35    | prof1      |
| EGC301P   | OS_Lab         | 192   | prof1      |
+-----+-----+-----+-----+
```

Report By:

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